

B. Tech. Examination, 2024-25

(Odd Semester)

MATRICES AND CALCULUS

Time : Three Hours]

[Maximum Marks : 60

Note :- Attempt all questions.

SECTION - A

1. Attempt all parts of the following : $8 \times 1 = 8$

• (a) Define skew hermitian and hermitian matrix.

✓ (b) For which value of k , the rank of the given matrix A is 3, where :

$$A = \begin{bmatrix} 2 & 4 & 2 \\ 3 & 1 & 2 \\ 1 & 0 & k \end{bmatrix}$$

(c) Find the n^{th} derivative of $\log x$.

[P. T. O.]

- (d) State Euler's theorem for homogeneous functions.

- (e) Evaluate:

$$\int_0^1 \int_{y^2}^{2-y} xy \, dx \, dy$$

- (f) Write the value of $\sqrt{1/2}$.

- (g) If $\vec{r} = \sin t \hat{i} + \cos t \hat{j} + t \hat{k}$ find:

$$\left| \frac{d^2 \vec{r}}{dt^2} \right|$$

- (h) State Gauss divergence theorem.

SECTION - B

2. Attempt any two parts of the following: $2 \times 6 = 12$

- (a) Verify the Cayley-Hamilton theorem for the matrix:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$$

Also find its inverse by using this theorem.

- (b) If $y = \cos(m \sin^{-1} x)$ then prove that:

$$(1 - x^2) y_{n+2} - (2n + 1) x y_{n+1} + (m^2 - n^2) y_n = 0$$

- (c) Prove that:

$$2^{2n-1} \sqrt{n} \sqrt{n + 1/2} = \sqrt{\pi} \sqrt{2n}$$

- (d) Show that:

$$\int \int_S \vec{F} \cdot \hat{n} \, dS = \frac{3}{2}$$

where

$$\vec{F} = 4xz \hat{i} - y^2 \hat{j} + yz \hat{k}$$

and S is the surface of the cube bounded by the planes $x=0, x=1, y=0, y=1, z=0, z=1$.

SECTION - C

Note :- Attempt all questions. Attempt any two parts from each questions. $8 \times 5 = 40$

3. (a) Find the rank of the matrix by reducing it into normal form:

$$\begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix}$$

[P. T. O.]

- (b) Determine k such that the system of homogeneous equations :

$$2x + y + 2z = 0$$

$$x + y + 3z = 0$$

$$4x + 3y + kz = 0$$

has (i) Trivial solution

(ii) Non-trivial solution

- (c) Find the eigen-values and corresponding eigen-vectors of the matrix :

$$\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$$

4. (a) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show that :

$$\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z} \right)^2 u = \frac{-9}{(x+y+z)^2}$$

- (b) Expand $e^x \sin y$ in powers of x and y as far as terms of third degree.

- (c) Discuss the maximum and minimum of :

$$x^2 + y^2 + 6x + 12$$

5. (a) Change the order of integration in :

$$I = \int_0^1 \int_{x^2}^{2-x} f(x, y) dy dx$$

- (b) Find the area between parabolas $y^2 = 4ax$ and $x^2 = 4ay$.

- (c) Evaluate :

$$\int_0^1 \int_1^2 \int_2^3 (x+y+z) dx dy dz$$

6. (a) A particle moves on the curve $x = 2t^2$, $y = t^2 - 4t$, $z = 3t - 5$, where t is the time. Find the component of velocity and acceleration at time $t = 1$ in the direction $\hat{i} - 3\hat{j} + 2\hat{k}$.

- (b) Prove that :

$$(y^2 - z^2 + 3yz - 2x)\hat{i} + (3xz + 2xy)\hat{j} + (3xy - 2xz + 2z)\hat{k}$$

is both solenoidal and irrotational.

- (c) Find the total work done by a force :

$$\vec{F} = (x^2 + y^2)\hat{i} - 2xy\hat{j}$$

in moving a point from $(0, 0)$ to (a, b) along the rectangle bounded by the lines $x = 0$, $x = a$, $y = 0$ and $y = b$.
